

Sprint 1 Bootstrap: 36-State Factored MDP with LLM-Estimated Transitions

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July 2026

1 MDP Formulation

The Sprint 1 simulator implements a finite-horizon factored MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$ with 36 states, 4 actions, and transitions estimated by an LLM (deepseek-v4-flash) using Algorithm 2.

1.1 State Space

Four factors: step bin (`step_bin` $\in \{\text{inactive}, \text{moderate}, \text{active}\}$), sleep quality (`sleep` $\in \{\text{good}, \text{poor}\}$), day type (`day_of_week` $\in \{\text{weekday}, \text{weekend}\}$), and burden level (`burden` $\in \{\text{low}, \text{medium}, \text{high}\}$):

$$\mathcal{S} = \{\text{inactive}, \text{moderate}, \text{active}\} \times \{\text{good}, \text{poor}\} \times \{\text{weekday}, \text{weekend}\} \times \{\text{low}, \text{medium}, \text{high}\}$$

Three factors are deterministic: `day_of_week` cycles daily through a 7-day pattern; `burden` uses a rolling window of the last 3 actions (increments on non-idle actions). Two factors are stochastic: `step_bin` at each timestep, and `sleep` at day boundaries.

1.2 Action Space

$$\mathcal{A} = \{\text{idle}, \text{movement_suggestion}, \text{goal_reminder}, \text{journal}\}$$

All non-idle actions incur a penalty of 0.05.

1.3 Transition Function

Transitions are loaded from 6 JSON probability tables generated by an LLM (deepseek-v4-flash, $N = 16$ samples per state-action pair, Algorithm 2):

- `day_boundary.json`: $P(\text{sleep}' \mid \text{step_bin}, \text{burden}, \text{day_of_week}, \text{sleep})$ – action-independent, evaluated at step 0 of each day.
- `within_day_0.json` through `within_day_4.json`: $P(\text{step_bin}' \mid \text{step_bin}, \text{burden}, \text{action}, \text{day_of_week}, \text{sleep})$ – one table per timestep, action-conditioned.

The LLM’s estimates reflect plausible real-world dynamics: interventions have modest effects on step counts, and sleep quality influences next-day activity.

1.4 Reward

$$R = \alpha \cdot \text{step_bin_value} + (1 - \alpha) \cdot \text{sleep_value} - \text{action_penalty}$$

with $\alpha = 0.9$. Step bin values: `inactive` $\rightarrow$ 0.0, `moderate` $\rightarrow$ 0.5, `active` $\rightarrow$ 1.0. Sleep values: `good` $\rightarrow$ 1.0, `poor` $\rightarrow$ -1.0. Action penalty: 0.0 for idle, 0.05 otherwise.

Optional `reward_multiplier_by_step` masks the last step of each day: [1, 1, 1, 1, 0].

1.5 Episode

$T = 90 \text{ days} \times 5 \text{ steps/day} = 450 \text{ timesteps per episode}$.

2 Reference Policy Bounds

Six reference policies evaluated via simulation (10 seeds, 450 timesteps):

Policy	Total Reward	Per Step	Description
Fixed idle	380.4 ± 3.7	0.845	Never intervene (no penalty)
Greedy oracle (1-step)	281.0 ± 20.9	0.625	Myopic expected-reward maximisation
Random	100.1 ± 11.7	0.222	Uniform action selection
Fixed movement_suggestion	35.2 ± 4.8	0.078	Always suggest movement
Fixed journal	50.5 ± 5.1	0.112	Always journal
Fixed goal_reminder	91.2 ± 7.1	0.203	Always remind of goal

Table 1: Reference policy bounds (10 seeds). Always-idle dominates because the LLM estimates natural activity is already high and action penalties are costly. The greedy oracle’s myopic one-step lookahead fails to capture long-term value.

Always-idle at 0.845/step sets an upper bound for agents: the LLM estimates that most users maintain moderate-to-active step counts without intervention, and paying a 0.05 penalty for no improvement is worse than doing nothing.

3 Base Benchmark Results

50 seeds, 450 timesteps, default hyperparameters (EG $\epsilon = 0.1$, UCB $c = 2.0$, TS $\alpha_0 = \beta_0 = 1$). Config: `configs/sprint1_bootstrap.yaml`.

Agent	Total Reward	Per Step	Last 50 Steps
Epsilon-Greedy ($\epsilon = 0.1$)	280.8 ± 89.3	0.624	0.743
Thompson Sampling ($\alpha_0 = \beta_0 = 1$)	252.6 ± 105.6	0.561	0.662
UCB ($c = 2.0$)	240.4 ± 33.9	0.534	0.803
Random	99.7 ± 10.9	0.222	0.215

Table 2: Base benchmark results (50 seeds, 450 timesteps). Learning agents massively outperform random (2.5–2.8 $\times$), confirming the bootstrap tables contain exploitable structure. EG leads.

Key observation: under bootstrap transitions, learning agents achieve 2.5–2.8 $\times$ random, starkly different from the random-transition baseline where all agents clustered within 3% of each other. The LLM-estimated dynamics have meaningful structure.

4 Full Benchmark Results

50 seeds, 450 timesteps, 9 agents with MVP-optimised hyperparameters (EG $\epsilon = 0.05$, D-EG $\epsilon_{\text{start}} = 0.2$, $T_{\text{decay}} = 200$, UCB $c = 0.5$). Config: `configs/sprint1_bootstrap_extensions.yaml`.

Agent	Total Reward	Per Step	Last 50 Steps
Standard UCB ($c = 0.5$)	358.1 ± 14.0	0.796	0.837
Standard D-EG	287.8 ± 106.9	0.640	0.718
Standard EG ($\epsilon = 0.05$)	282.3 ± 95.0	0.627	0.748
Contextual UCB	268.3 ± 29.6	0.596	0.733
Standard TS	252.6 ± 105.6	0.561	0.662
Contextual D-EG	240.4 ± 93.0	0.534	0.614
Contextual EG	216.9 ± 95.7	0.482	0.583
Contextual TS	155.5 ± 70.8	0.346	0.399
Random	99.7 ± 10.9	0.222	0.215
Fixed idle (upper bound)	380.4	0.845	0.845

Table 3: Full benchmark results (50 seeds). Standard UCB at $c = 0.5$ reaches 94% of the always-idle ceiling. Contextual agents universally underperform their standard counterparts.

Key findings:

1. **Std UCB at $c = 0.5$ is dominant:** 358.1 total, 94% of the always-idle ceiling, with the lowest variance (14.0 vs 70–106 for others). Hyperparameter tuning is critical: the default $c = 2.0$ yielded only 240.4.
2. **Contextual agents are worse:** Every contextual variant underperforms its standard counterpart, the opposite of the MVP result. This suggests `step_bin` as a context feature misleads agents in the bootstrap dynamics – transitions are non-stationary (vary by timestep) and the current `step_bin` does not reliably predict intervention efficacy.
3. **The $c = 0.5$ UCB achieves consistency:** with ± 14.0 std dev vs ± 33.9 at $c = 2.0$, tuning simultaneously improves both mean and variance.

5 Masked Reward Benchmark

Same 9 agents on `configs/sprint1_bootstrap_extensions_masked.yaml` where step 4 (end of day) yields zero reward. Results are 50 seeds.

6 Comparison to Random-Transition Baseline

7 Context Feature Comparison

The extensions benchmark used `step_bin` as the context feature (3 bins, same fragmentation as `step_bin`). To test whether a better-chosen context could close the gap, we swapped to `burden` (3 levels, same fragmentation) and re-ran the 9-agent benchmark. Results are 50 seeds.

Despite having $13\times$ higher ANOVA F-statistic as a predictor of next-step activity, `burden` makes a worse context feature. The reason is that `burden` is *endogenous* – it is a rolling window count

Agent	Total Reward	Per Step	Last 50 Steps
Standard UCB ($c = 0.5$)	265.9 ± 37.3	0.591	0.681
Standard D-EG	248.3 ± 76.2	0.552	0.616
Standard EG ($\epsilon = 0.05$)	195.8 ± 98.4	0.435	0.518
Contextual UCB	178.4 ± 35.2	0.397	0.501
Standard TS	174.6 ± 70.6	0.388	0.456
Contextual D-EG	169.9 ± 70.0	0.378	0.435
Contextual EG	163.7 ± 73.9	0.364	0.443
Contextual TS	119.0 ± 38.5	0.264	0.289
Random	83.6 ± 8.4	0.186	0.179

Table 4: Masked benchmark results (50 seeds, step 4 zero-reward). Relative ordering is preserved. D-EG is most mask-resilient (86% retention).

Metric	Random Transitions	Bootstrap Transitions
Best agent	Std D-EG 189.1 (0.420)	Std UCB 358.1 (0.796)
Random baseline	183.5 (0.408)	99.7 (0.222)
Best vs random gap	3%	259%
Contextual advantage	None ($\approx 0\%$)	Negative (all worse)
Variance (best agent)	± 82	± 14

Table 5: Bootstrap transitions vs random transitions. The LLM-estimated tables produce structured dynamics with large agent differentials, while random transitions wash out all policy differences. Std UCB’s low variance under bootstrap suggests reliable convergence.

of the agent’s past non-idle actions. When a fragmented contextual agent explores, it takes non-idle actions, which increase burden, which shifts the context distribution, which triggers further exploration. This creates a feedback loop:

exploration $\rightarrow$ higher burden $\rightarrow$ unfamiliar context $\rightarrow$ more exploration $\rightarrow \dots$

In contrast, `step_bin` is at least partly exogenous – it depends on the LLM-estimated transition dynamics and is not directly controlled by recent actions. The `step_bin` distribution is relatively stable regardless of the agent’s policy, making it a safer (if less predictive) context feature.

Both context features are dominated by the standard (non-contextual) UCB, confirming that `step_bin` context does not help this MDP.

8 Horizon Scaling

To test whether contextual agents would catch the standard UCB given more data, we re-ran both context-feature benchmarks at 900 days (4500 steps, $10\times$) and 9000 days (45000 steps, $100\times$). All other settings are identical (50 seeds, same hyperparameters).

Several patterns emerge:

1. **Ctx UCB (`step_bin`) effectively closes the gap.** At 90d / 450 steps it achieves 75% of Std UCB’s total; at 900d / 4500 steps it reaches 95%; at 9000d / 45000 steps it reaches 99.3% (37971.9 vs 38241.5). The contextual fragmentation penalty is purely a *sample-efficiency*

Agent	No Context	Ctx <code>step_bin</code>	Ctx <code>burden</code>
Standard UCB ($c = 0.5$)	358.1 ± 14.0	—	—
Contextual UCB	—	268.3 ± 29.6	250.6 ± 39.7
Contextual D-EG	—	240.4 ± 93.0	216.0 ± 77.6
Contextual EG	—	216.9 ± 95.7	200.4 ± 82.9
Contextual TS	—	155.5 ± 70.8	158.3 ± 56.1
Random	99.7 ± 10.9	—	—

Table 6: Comparison of context features (50 seeds). Burden context underperforms `step_bin` context across all agents except TS (where the difference is negligible).

Agent	90d / 450 st.	900d / 4500 st.	9000d / 45000 st.	Per-step
Std UCB ($c = 0.5$)	358.1 ± 14.0	3795.7 ± 24.0	38241.5 ± 51.1	0.8498
Ctx UCB (<code>step_bin</code>)	268.3 ± 29.6	3606.6 ± 43.5	37971.9 ± 63.9	0.8438
Ctx UCB (<code>burden</code>)	250.6 ± 39.7	3389.9 ± 79.5	36404.1 ± 374.9	0.8090
Std EG ($\epsilon = 0.05$)	282.3 ± 95.0	3578.6 ± 264.7	37182.9 ± 264.0	0.8263
Ctx EG (<code>step_bin</code>)	216.9 ± 95.7	3376.6 ± 423.0	36824.1 ± 952.6	0.8183
Std D-EG	287.8 ± 106.9	3373.3 ± 916.3	37097.9 ± 2987.5	0.8244
Ctx D-EG (<code>step_bin</code>)	240.4 ± 93.0	3168.0 ± 763.3	37057.1 ± 1590.5	0.8235
Std TS	252.6 ± 105.6	3354.3 ± 842.0	37060.1 ± 3511.0	0.8236
Ctx EG (<code>burden</code>)	200.4 ± 82.9	3140.1 ± 386.4	35008.6 ± 1722.7	0.7780
Ctx D-EG (<code>burden</code>)	216.0 ± 77.6	2694.5 ± 927.8	31957.4 ± 6113.1	0.7102
Ctx TS (<code>step_bin</code>)	155.5 ± 70.8	2215.5 ± 737.6	25537.2 ± 8989.0	0.5675
Ctx TS (<code>burden</code>)	158.3 ± 56.1	1776.2 ± 571.3	19794.5 ± 5612.9	0.4399
Random	99.7 ± 10.9	1020.8 ± 45.1	10177.4 ± 118.4	0.2262

Table 7: Horizon scaling: 90d, 900d, and 9000d (50 seeds). At 9000d Ctx UCB (`step_bin`) reaches 99.3% of Std UCB. Rank ordering is stable across all three horizons.

cost, not an asymptotic limitation. Given enough data, `step_bin`-contextual UCB converges to the same optimal policy of doing nothing.

2. **Ctx UCB (`burden`) also converges, but slower.** Even at 45000 steps, `burden`-context UCB reaches only 0.8090/step vs Std UCB’s 0.8498 (95.2%). The endogenous feedback loop does not disappear with more data, but its relative penalty shrinks: 75% $\rightarrow$ 89% $\rightarrow$ 95% as horizon increases.
3. **The 10 $\times$ multiplier gap disappears at longer horizons.** At 900d, contextual agents had higher gain multipliers (13–16 $\times$) than standard ones (11–13 $\times$). At 9000d the per-step rates show a different pattern: Std UCB converges to 0.8498, with most standard agents clustering at 0.823–0.826 and `step_bin`-contextual agents at 0.818–0.844. The gap is now driven by *asymptotic convergence rate* rather than a fixed fragmentation penalty.
4. **Rank ordering is remarkably stable across all three horizons.** No agent that was worse at 450 steps overtakes one that was better at 45000 steps. The relative hierarchy is fully determined early in training. However, some pairs converge: Ctx D-EG (`step_bin`)

nearly ties Std TS and Std D-EG at 45000 steps, suggesting the exploration decay eventually lets it converge.

5. **Std UCB at 45000 steps reaches 0.8498/step, matching the always-idle upper bound (0.845).** The standard deviation shrinks to ± 51.1 , the lowest of any agent at any horizon. UCB with $c = 0.5$ has converged to the optimal deterministic policy of always choosing idle.
6. **Contextual TS remains terrible even at 45000 steps.** Ctx TS (`step_bin`) at 0.5675/step and Ctx TS (`burden`) at 0.4399/step are the only agents that do not converge toward the standard-agent cluster. Beta-Bernoulli Thompson Sampling with independent per-context Beta posteriors cannot aggregate information across contexts, so it never escapes the fragmentation penalty.
7. **`step_bin` beats `burden` at all three horizons.** The gap between `step_bin` and `burden` UCB narrows from 17.7 (90d) to 216.7 (900d) to 1567.8 (9000d), but in relative terms `burden` per-step stays at 95–96% of `step_bin` per-step at all three horizons. The relative penalty is constant.

9 Key Findings

1. **Bootstrap transitions have exploitable structure.** Unlike the Dirichlet-random baselines where all agents performed similarly (~ 185 total), the LLM-estimated tables yield a 259% gap between best learner and random. The LLM produces non-trivial transition dynamics.
2. **Always-idle is near-optimal.** The LLM estimates that most users naturally maintain moderate-to-active step counts. Interventions incur a penalty without consistently improving outcomes, making inaction the best policy. Std UCB learns this implicitly, reaching 94% of the idle ceiling.
3. **Context features hurt, even with better signal.** Swapping from `step_bin` to `burden` – a context with $13\times$ stronger predictive power – made performance worse, not better. The reason is that `burden` is endogenous: exploration changes `burden`, which shifts the context distribution, which triggers more exploration. This feedback loop amplifies the sample-efficiency cost of context fragmentation.
4. **UCB hyperparameter sensitivity is extreme.** Default $c = 2.0$: 240. Tuned $c = 0.5$: 358. The MVP-optimised values transferred well, but a dedicated grid search may find even better settings.
5. **The greedy oracle (one-step lookahead) is not a good bound** at 281, far below Std UCB’s 358. Myopic expected-reward maximisation fails because actions affect future states through `burden` accumulation and sleep at day boundaries.
6. **The LLM’s transition estimates are a plausible starting point** but should be validated against real data. The strong idle preference may reflect LLM bias toward inaction rather than true user dynamics.